

HEFEIFEI JIANG

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RIVERSIDE, CA

EDUCATION

University of California, Riverside (UCR)

M.S. | *Robotics*

Sep 2025 - Present

Riverside, CA, USA

Shaanxi University of Science and Technology (SUST)

B.Eng. (First-class degree) | *Computer Science and Technology*

Sep 2020 - Jun 2024

Xi'an, China

Joint-Degree Undergraduate Program with Ulster University, UK

RESEARCH INTEREST

My research centers on **human-robot interaction**: how robots and people can work together effectively in shared physical spaces. I am interested in robots that involve humans in the loop, whether by taking guidance and corrections from people, assisting and following them, or adapting to their individual preferences, and in what makes such interaction natural, safe, and low-effort for the human. I approach these questions through **robot learning** and **manipulation**, using learning-based methods to enable robots to act with and around people. My goal is to build interactive robots that people find intuitive to guide, trust, and collaborate with.

PUBLICATIONS & PREPRINTS

* Equal contribution.

[6] Mingxuan Yan, Litian Gong, Dihong Huang, Zhixu Li, Tianyu Zhang, Wenqian Zhang, **Hefeifei Jiang**, Peihao Li, Jian Zhang, Mengfei Zhao, Yikai Tang, Hai Zhai, Zehao Wang, Ruijian Liang, Yanjia Huang, Lin Shao, Changliu Liu, Kaiyu Hang, Zhiwen Fan, Masayoshi Tomizuka, Jianfei Yang, Jiachen Li, “What Matters in Quality-Aware VLA Pretraining: A Benchmark and Empirical Study”, *IEEE International Conference on Robotics and Automation (ICRA)*, 2027. **Under review**.

[5] Shiting Gong, Jianpeng Yao, Litian Gong, **Hefeifei Jiang**, Huaide Jiang, Jinfeng Wang, Zheyu Lin, Mingyu Ding, Jiachen Li, “Closed-Loop Customizable Human Following via Distilling Monotonic Constrained Policies”, *IEEE International Conference on Robotics and Automation (ICRA)*, 2027. **Under review**.

[4] Zhefei Gong*, **Hefeifei Jiang***, Zehao Wang, Jiachen Li, “HiRe-MoMa: Hierarchical Residual Policy Learning for Real-World Mobile Manipulation with Sparse Human Guidance”, *Conference on Robot Learning (CoRL)*, 2026.

[3] Zehao Wang*, **Hefeifei Jiang***, Siyan Li, Guoyuan Wu, Jiachen Li, “CoopLight: Cooperative Vision-Language Agents for Long-Tail Traffic Control”, *Conference on Neural Information Processing Systems (NeurIPS)*, 2026. **Under review**.

[2] Y. Qi, M. Qiu, **H. Jiang**, F. Wang, “Extracting Fingerprint Features Using Autoencoder Networks for Gender Classification”, *Applied Sciences*, 12(19): 10152, 2022.

[1] Y. Qi, **H. Jiang**, S. Li, J. Cao, “ConvLSTM Coupled Economics Indicators Quantitative Trading Decision Model”, *Symmetry*, 14(9): 1896, 2022.

PATENTS

[2] Y. Qi, **H. Jiang**, F. Wang, J. Yang, “A Method and Apparatus for Matching a Three-dimensional Model of the Sacrum Based on Deep Learning”, Chinese patent.

[1] Y. Qi, **H. Jiang**, X. Jin, J. Ma, M. Zhao, Y. Liu, K. Qian, Y. Wang, Z. Wang, H. Cui, “AIOT Digital Twin Intelligent Substation Cloud-based Analysis Platform”, Chinese patent.

RESEARCH EXPERIENCE

Trustworthy Autonomous Systems Lab (TASL)

Advisor: Prof. Jiachen Li, UC Riverside

2025 - Present

Riverside, CA, USA

- **HiRe-MoMa: Real-World Fine-Tuning of Mobile Manipulation with Sparse Human Guidance (Co-first author)**
 - ▶ Built an asynchronous residual RL framework that fine-tunes frozen base policies (Diffusion Policy, ACT) on a TIAGo++ mobile manipulator with only 25–40 min of online interaction.
 - ▶ Designed hierarchical residual heads with unidirectional base-to-arm conditioning, and an expert-guided value alignment loss that uses under 20% human interventions to shape the critic.

- ▶ Raised real-world success on placement from 50% to 95% and on sweeping from 45% to 90%; paper accepted to CoRL 2026.
- **CoopLight: Cooperative Vision-Language Agents for Long-Tail Traffic Control (Co-first author)**
 - ▶ Developed a VLM-based framework where traffic-signal and vehicle agents perform confidence-aware VQA over multi-view images and exchange semantic reports for joint decisions under incidents, jaywalking, and emergency vehicles.
 - ▶ Built closed-loop SUMO and Blender simulations of two real intersections and a 7,828-pair VQA dataset with Qwen2.5-VL-3B agents; cut average waiting time by 26% overall and 56% for connected vehicles; paper under review at NeurIPS 2026.
- **Closed-Loop Customizable Human Following via Distilling Monotonic Constrained Policies (Co-author)**
 - ▶ Distilled constrained-RL guiding policies into a preference-conditioned meta-policy with closed-loop adaptive mapping, enabling test-time following-distance adjustment of up to 114% without retraining.
 - ▶ Deployed on a ROSMASTER X3 robot with LiDAR-based pedestrian tracking, showing sim-to-real transfer indoors and outdoors; paper under review at ICRA 2027.

Quantum Neural Networks for Efficient Multimodal Representation Learning

Jun 2024 - Dec 2024

Remote Research Intern, Advisor: Prof. Juan Zhuang

Remote

- Replaced the CLIP image encoder with a Quantum Neural Network (QNN) to shrink model size while preserving multimodal representation performance.
- Designed and implemented a QNN-based visual encoder, integrated it into a CLIP-style framework, and evaluated on benchmark datasets with competitive accuracy at a substantially reduced parameter budget.

Summer Workshop at National University of Singapore

Jun 2024 - Jul 2024

Team Leader, Advisor: Prof. Tan Wee Kek

Singapore

- Built a smart greenhouse monitoring system with Raspberry Pi, ESP32, and multi-modal sensors (temperature, humidity, CO₂, soil moisture and temperature), plus a full-stack web app for real-time monitoring and pest detection.
- Streamed sensor readings and video via Flask APIs; applied YOLOv8l for potted-plant counting and a ResNet classifier for pest and disease recognition.
- Coordinated task decomposition and integration in a multinational team (hardware-software co-design).

Digital Twin Intelligent Grid Physical Fusion Platform

Apr 2021

Robot Embedded Development

China

- Developed a smart-grid monitoring framework combining digital twins, inspection robots, and cloud-based management software; deployed in multiple substations as a provincial-level innovation project with a software copyright.
- Built a tracked inspection robot with SLAM-based mapping, multi-sensor deployment, obstacle detection and automatic inspection; trained and deployed a YOLOv3 recognition module on electrical-utility datasets.

SKILLS

Programming & ML: Python, C/C++, PyTorch, Bash, Git, LaTeX

Robotics: ROS, TIAGo++, Franka, Diffusion Policy, ACT

Simulation & Benchmarks: IsaacLab, DROID, LIBERO, SUMO, Webots

Methods: Reinforcement Learning, Imitation Learning, Residual Policy Learning, Vision-Language Models

HONORS & AWARDS

International

- **Meritorious Award**, Mathematical Contest in Modeling (MCM) Feb 2022
- **Honorable Mention**, ICPC Asia-East Continent Final Apr 2021
- **Gold Medal**, International Festival of Science and Technology, Tunisia Aug 2020

Domestic

- **"Zhi Cheng Zhi Bo" All-Round Excellence Scholarship (Top 10)**, University-level Scholarship Jun 2024
- **Zhou Lian Scholarship (Top 10, #1 by points)**, University-level Scholarship Apr 2024
- **5th Place**, 4th "Keda" Cup Five-a-Side Football Tournament Jun 2024
- **3rd Prize**, National College Optoelectronic Design Competition Aug 2022
- **3rd Prize**, National College Computer Design Competition Aug 2022
- **Ranked 215 / 627**, WeChat Big Data Challenge May 2022
- **Excellence Award**, China AI+ Innovation and Entrepreneurship Competition Dec 2021
- **Provincial-Level Approval**, China College Student Innovation and Entrepreneurship Project Nov 2021
- **1st Prize**, 23rd National Robot Championship & 12th International Humanoid Robot Olympics Oct 2021
- **3rd Prize**, National College Computer Design Competition Aug 2021